ECE 554 Project Proposal

Team Members:

- David Eapen: RTL, RISC-V, Software Development, Algorithms, and Audio Nerd
- Kailan Kraft: RTL, Machine Learning, Computer Architecture
- Nate Parmley: Digital Signal Processing, RTL Design, Sensor Interfacing
- Sandeep Sankar: Computer Architecture, RTL Design, Assembly Testing
- Srivibhav Jonnalagadda: Computer Architecture, RTL Design, Scripting, Testbench Development

Project Overview:

This project aims to design and implement an FPGA-based acoustic camera capable of visualizing the spatial location and intensity of sound sources using a 64-microphone array and a USB camera feed. The system will allow users to select specific frequency bands (including ultrasonic frequencies) and amplitude ranges, and will overlay a sound intensity heatmap onto the live video feed.

The primary application of this system is detecting and locating gas leaks in indoor environments. Many compressed gas leaks emit ultrasonic sound that is inaudible to humans but would be detectable with our device. Gas leak detection is critical in industrial and laboratory settings, where even small leaks can cost thousands of dollars over time. Ultrasonic acoustic imaging provides a non-invasive and real-time method of identifying leak locations. Commercial acoustic cameras are often expensive and proprietary.

Technical Approach:

The system consists of two primary components:

1. External Hardware
    - 64-element MEMS microphone array (SPH0655LM4H-1-8 microphones)
    - USB camera module (OV2710-based 1080p camera)
    - Custom PCB for microphone layout and routing
    - Display to output camera feed with acoustic intensity overlay
2. FPGA System
    - Custom Out-of-Order RISC-V Processor
    - Hardware-accelerated signal processing blocks
    - Video overlay pipeline

Methodology:

The 64-microphone array will capture time-domain signals simultaneously. Localization will be achieved using phase difference across microphones and Time Difference of Arrival (TDOA). The primary processing steps will be parallel sampling of microphone signals, DFT computation, frequency filtering (user-selectable band), phase difference computation, and intensity heatmap generation. These steps require high-throughput parallel DSP operations, motivating hardware acceleration.

The FPGA design consists of three major RTL subsystems:

- *Out-of-Order RISC-V Processor:* We will design and synthesize a custom out-of-order RISC-V core that includes a reorder buffer (ROB), reservation stations, dynamic scheduling, and speculative execution. This processor will control system configuration, manage frequency selection and filtering parameters and coordinate accelerator blocks.

- *Custom Signal Processing Accelerators:* Dedicated RTL blocks will accelerate parallel DFT engines, phase extraction units and heatmap generation. These accelerators reduce computational latency compared to software-only execution on the RISC-V core.
- *Video Overlay Engine:* A hardware video pipeline will receive camera frame input, generate acoustic intensity map, and output real-time visualization.

Course Fit & Justification:
This project is appropriate for ECE 554 because we will be implementing our own custom RISC-V processor and using it to design a fairly novel technology that is the acoustic camera. Our design requires FPGA acceleration because signal processing across 64 channels in real time is computationally intensive. Custom logic is necessary to achieve parallel DFT processing and compute phase differences simultaneously. Also, our design will involve complex interfacing to achieve 64-microphone synchronization, high-bandwidth data movement and camera feed integration. This is a full-system RTL design with significant architectural depth.

Execution Strategy:
Due to the complexity of our design, we anticipate some challenges. These challenges include successfully synchronizing 64 microphone inputs, meeting FPGA resource and DSP block constraints, achieving timing closure at target clock frequency, managing video bandwidth and memory buffering, and preserving PCB signal integrity.

To mitigate road blocks during the project we will begin with an 8–16 microphone prototype before scaling to 64. Additionally, we will design accelerators modularly to allow scaling. If necessary, we may have to reduce microphone count or simplify out-of-order features. The system will be developed incrementally to ensure continuous demonstrable progress.

Project Timeline & Evaluation:
- *Midterm Milestones:* Functional microphone subset (8–16 channels). Working DFT accelerator. Baseline RISC-V processor operational. Demonstrated phase-based localization.
- *Final Milestones:* Full 64-channel acquisition. Functional out-of-order RISC-V core. Real-time heatmap overlay. Demonstration of ultrasonic source localization.

Development Timeline:

Week	Tasks
Mar 1-7	Complete architecture interface, PCB Schematic, Microphone interface RTL
Mar 8-14	Functional DFT accelerator, Baseline RISC-V pipeline
Mar 15-21	Functional signal processing accelerators, Extend RISC-V core for Out-of-Order execution
Mar 22-28	Complete PCB integration, Video overlay pipeline, Demonstrated localization with microphone subset
Mar 29 - Apr 4	Demonstrated localization with 64 microphone array, Functional heatmap overlay
Apr 5-11	Full system integration, Demonstrated ultrasonic localiztion with visual overlay
Apr 12-18	Optimization, Final validation

Team Responsibilities:
- David Eapen: Initial testing of sensor (with MCU), RTL for DFT accelerator, assist with RISC-V processor, soldering
- Kailan Kraft: RTL of Video Overlay Engine, assist with DSP functions and RISC-V if needed
- Nate Parmley: RTL Design of Signal Processing Accelerators, PCB Design and Integration
- Sandeep Sankar: RTL Design of OoO RISC-V core and accelerator blocks, RTL testing
- Srivibhav Jonnalagadda: RTL Design of OoO RISC-V core, Automation, and Testbench Design Architect

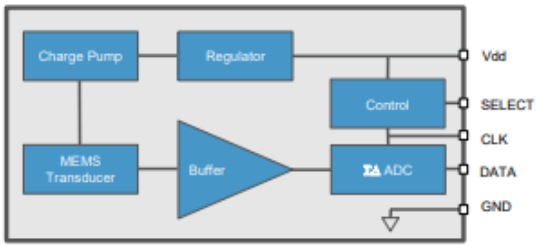
References

SPH0655LM4H-1-8 MEMS Digital Microphone Specification:

MULTIMODE DIGITAL BOTTOM PORT SISONIC™ MICROPHONE

knowles

The SPH0655LM4H-1 is a miniature, high-performance, low power, bottom port silicon digital microphone with a single bit PDM output. Using Knowles' proven high performance SiSonic™ MEMS technology, the SPH0655LM4H-1 consists of an acoustic sensor, a low noise input buffer, and a sigma-delta modulator. These devices are suitable for applications such as cellphones, smart phones, laptop computers, sensors, digital still cameras, portable music recorders, and other portable electronic devices where excellent wideband audio performance and RF immunity are required. In addition, the SPH0655LM4H-1 offers multiple performance modes.

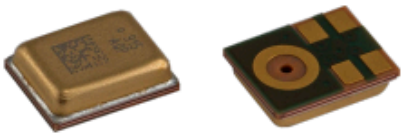


ABSOLUTE MAXIMUM RATINGS

Table 1: Absolute Maximum Ratings

Parameter	Absolute Maximum Rating	Units
Vdd to Ground	-0.5, +5.0	V
DATA, CLOCK, SELECT to Ground	-0.3, +5.0	V
Input Current	±5	mA
Short Circuit to/from DATA	Indefinite to Ground or Vdd	sec
Storage Temperature	-40 to +100	°C
Operating Temperature	-40 to +100	°C

Stresses exceeding these "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only. Functional operation at these or any other conditions beyond those indicated under "Acoustic & Electrical Specifications" is not implied. Exposure beyond those indicated under "Acoustic & Electrical Specifications" for extended periods may affect device reliability.



PRODUCT FEATURES

- Low Distortion /High AOP
- High SNR
- Low Current Consumption in Low-Power Mode
- Flat Frequency Response
- High Drive Capability
- RF Shielded
- Bottom Port
- Sensitivity Matching
- Supports Dual Multiplexed Channels
- Multiple Performance Modes (Sleep, Low-Power, Normal)
- Ultra-Stable Performance
- Omnidirectional
- Standard SMD Reflow
- LGA Package

TYPICAL APPLICATIONS

- Portable Electronics
- Cellphones
- Laptop Computers
- Tablets
- Digital Still Cameras
- Portable Music Recorders

ACOUSTIC & ELECTRICAL SPECIFICATIONS[1]

Table 2: General Microphone Specifications

Test Conditions: 23 ±2°C, 55±20% R.H., Vdd=1.8 V, Tedge ≤ 3ns, unless otherwise indicated

Parameter	Symbol	Conditions	Min	Typ	Max	Units
Supply Voltage	Vdd		1.65	1.8	3.6	V
Low Frequency Rolloff	LFRO	-3dB relative to 1 kHz	-	25	-	Hz
High Frequency Flatness		+3dB relative to 1 kHz	-	22	-	kHz
Resonant Frequency Peak	Fres	Free Field response	-	36	-	kHz
DC Offset		Fullscale = ±100	-3.25	-	0.1	% FS
Directivity			Omnidirectional			
Polarity		Increasing sound pressure	Increasing density of 1's			
Data Format			½ Cycle PDM			
Sensitivity Drop		Vdd(min) ≤ Vdd ≤ Vdd(max)	-	-	±0.25	dB
Clock Input Capacitance	Cin		-	15	-	pF
Data Output Capacitance	Cout		-	55	-	pF
Data Output Load	Cload		-	-	110	pF
SELECT (high)			Vdd-0.2	-	3.6	V
SELECT (low)			-0.3	-	0.2	V
Short Circuit Current	Isc	Grounded DATA pin	1	-	20	mA
Fall-asleep Time[3,4]		Fclock < 250kHz	-	-	10	ms
Wake-up Time[3,5]		Fclock ≥ 500kHz	-	-	35	ms
Startup Time[3]		Powered Down → Active, S within 1 dB of final value	-	-	35	ms
Time to First Data Bit[6]		Time from valid Vdd and CLK until the first logical bit is driven on the DATA line. The output is tristate until First Data Bit.	-	4	-	ms
Mode-Change Time[3, 6]		Low Power Mode ⇔ Normal Mode	-	-	17	ms

Table 3: Normal Mode

Test Conditions: 23 ±2°C, 55±20% R.H., Vdd=1.8 V, Fclock = 2.4 MHz (D.C. = 50%), Tedge ≤ 3ns, SELECT grounded, no load, unless otherwise indicated

Parameter	Symbol	Conditions	Min	Typ	Max	Units
Supply Current[2]	Idd	Fclock = 1.2 MHz	-	650	690	µA
		Fclock = 1.536 MHz	-	750	820	
		Fclock = 2.4 MHz	800	1000	1200	
		Fclock = 3.072 MHz	-	1225	1350	
		Fclock = 4.8 MHz	-	1850	2000	
Sensitivity	S	94 dB SPL @ 1 kHz	-38	-37	-36	dBFS
Signal to Noise Ratio	SNR	94 dB SPL @ 1 kHz, A-weighted, Fclock = 1.2 MHz	-	64.5	-	dB(A)
		94 dB SPL @ 1 kHz, A-weighted, Fclock = 1.536 MHz	-	66	-	
		94 dB SPL @ 1 kHz, A-weighted, Fclock = 2.4 MHz	-	66	-	
		94 dB SPL @ 1 kHz, A-weighted, Fclock = 3.072 MHz	-	66	-	
		94 dB SPL @ 1 kHz, A-weighted, Fclock = 4.8 MHz	-	66	-	
Near-Ultrasonic SNR		94 dB SPL, @ 19 kHz , BW = 18.5 - 20.0 kHz	-	78	-	dB
Total Harmonic Distortion	THD	94 dB SPL @ 1 kHz	-	0.05	-	%
		115 dB SPL @ 1 kHz	-	0.2	-	
		1% THD @ 1 kHz, S = typ	-	130.5	-	dB SPL
Acoustic Overload Point	AOP	10% THD @ 1 kHz, S = typ	-	132.5	-	dB SPL
Power Supply Rejection Ratio	PSRR	200 mVpp sinewave @ 1 kHz	-	86	-	dB V/FS
Power Supply Rejection	PSR+N	200 mVpp 7/8 duty cycle rectangular waveform @ 217 Hz, A-weighted	-	-98	-	dBFS(A)

Table 4: Low-Power Mode

Test Conditions: 23 ±2°C, 55±20% R.H., Vdd=1.8 V, Fclock = 768 kHz (D.C. = 50%), Tedge ≤ 3ns, SELECT grounded, no load, unless otherwise indicated

Parameter	Symbol	Conditions	Min	Typ	Max	Units
Supply Current[2]	Idd		240	260	330	µA
Sensitivity	S	94 dB SPL @ 1 kHz	-22	-21	-20	dBFS
Signal to Noise Ratio	SNR	94 dB SPL @ 1 kHz, A-weighted (BW = 8 kHz)	-	64.5	-	dB(A)
Total Harmonic Distortion	THD	94 dB SPL @ 1 kHz	-	0.05	-	%
		1% THD @ 1 kHz, S = typ	-	113	-	dB SPL
Acoustic Overload Point	AOP	10% THD @ 1 kHz, S = typ	-	116	-	dB SPL
Power Supply Rejection Ratio	PSRR	200 mVpp sinewave @ 1 kHz	-	75	-	dBV/FS
Power Supply Rejection	PSR+N	200 mVpp 7/8 duty cycle rectangular waveform @ 217 Hz, A-weighted	-	-85	-	dBFS(A)

Table 5: Sleep Mode

Test Conditions: 23 ±2°C, 55±20% R.H., Vdd=1.8 V, Fclock = 0 Hz, SELECT grounded, no load, unless otherwise indicated

Parameter	Symbol	Conditions	Min	Typ	Max	Units
Sleep Current	Isleep		-	40	55	µA

[1] Sensitivity and Supply Current are 100% tested.

[2] Idd varies with Cload according to: ΔIdd = 0.5*Vdd*ΔCload*Fclock.

[3] Valid microphones states are: Powered Down Mode (mic off), Sleep Mode (low current, DATA = high-Z, fast startup), Low-Power Mode (low clock speed) and Normal Mode.

[4] Time from Fclock < 250 kHz to Isleep specification is met when transitioning from Active Mode to Sleep Mode.

[5] Time from Fclock ≥ 500 kHz to all applicable specifications are met when transitioning from Sleep Mode to Active Mode.

[6] Audio is temporarily muted during the transition between any microphone state.

Table 6: Digital Interface

Test Conditions: 23 ±2°C, 55±20% R.H., Vdd=1.8 V, Tedge ≤ 3ns, unless otherwise indicated

Parameter	Symbol	Conditions	Min	Typ	Max	Units
Logic Input High[7]	Vih		0.65xVdd	-	3.6	V
Logic Input Low[7]	Vil		-0.3	-	0.35xVdd	V
Logic Output High[7]	Voh	I_{OUT} = 2 mA	Vdd-0.45	-	Vdd	V
Logic Output Low[7]	Vol	I_{OUT} = 2 mA	0	-	0.45	V
Low→High Threshold[8]	Vl-h		-	-	0.65xVdd	V
High→Low Threshold[8]	Vh-l		0.35xVdd	-	-	V
Hysteresis Width[8]	Vhyst		0.05xVdd	-	0.20xVdd	V
Clock Frequency[7]	Fclock	Sleep Mode	0	-	250	kHz
		Low-Power Mode	500	-	900	kHz
		Normal Mode	1.1	-	4.8	MHz
Clock Duty Cycle	D.C.		40	50	60	%
Delay Time to Data Line Driven[7]	Tdd		18	-	35	ns
Delay Time to Valid Data[7]	Tdv	Max Cload	-	-	100	ns
Delay Time to High Z[7]	Tdz		5	-	16	ns
Hold Time[7]	Thold	Thold, as observed by the input device, will be dependent on Cload	5	-	-	ns

[7] See Figure 1: Timing Diagram.

[8] See Figure 2: Hysteresis Diagram.

ELP-USBFHD01M 1080P USB Camera Module Specification:

Specification:

Model	ELP-USBFHD01M-L21 (2.1mm lens) ELP-USBFHD01M-L36 (3.6mm lens) ELP-USBFHD01M-L60 (6mm lens) ELP-USBFHD01M-L80 (8mm lens) ELP-USBFHD01M-L12 (12mm lens)
Sensor	OV2710
Lens Size	1/2.7 inches
Pixel Size	3.0um X 3.0um
image area	5856 μm x 3276 μm
Max. Resolution	FULL HD 1920(H)X1080(V)
Compression format	MJPEG / YUV2 (YUYV)
Resolution & frame	320X240 QVGA MJPEG @100fps/ 352X288 CIF MJPEG @100fps 640X480 VGA MJPEG@100fps/ 800X600 SVGA MJPEG@60fps 1024X768 XGA MJPEG@30fps/ 1280X720 HD MJPEG@60fps 1280X1024 SXGA MJPEG@30fps/ 1920X1080 FHD MJPEG@30fps
Special function	Lens correction/ Defective pixel correction/ Black sun cancellation
Mini illumination	0.05lux
Shutter Type	Electronic rolling shutter / Frame exposure
Sensitivity	3.7V/lux-sec@550nm
Connecting Port type	USB2.0 High Speed (Support USB2.0 HS/FS, USB2.0 OTG)
USB Protocol	USB2.0 HS/FS USB1.1/USB1.0
Connecting Port type	USB2.0 High Speed
OTG Protocol	USB2.0 OTG
Free Drive Protocol	USB Video Class (UVC)
AEC	Support
AEB	Support
AGC	Support
Adjustable parameters	Brightness, Contrast, Saturation, Hue, Sharpness, Gamma,White balance, Backlight Contrast, Exposure
Lens Parameter	M12 2.1mm lens Optional 2.8mm/3.6mm/6mm/ FOV(D)170 Degree
Night vision	Support, need to equipped IR sensor 850nm or 940nm lens and IR LED Board
LED board power connector	Support 2P-2.0mm socket
Power supply	USB BUS POWER 4P-2.0mm socket
Operating Voltage	DC5V
Working current	120mA~220mA
Working temperature	0 ~ 60°C
Storage temperature	-20 ~ 75°C
Board size /Weight	38X38mm (Compatible 32X32mm) / about 30g
Cable	Standard 1M (optional 2M,3M,5M)
Operating system request	WinXP/Vista/Win7/Win8 Linux with UVC (above linux-2.6.26) MAC-OS X 10.4.8 or later Wince with UVC Android 4.0 or above with UVC